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Chapter 5

Differentiation

Any man who can drive safely while kissing a pretty girl is simply not giving the kiss the attention it deserves.

—Albert Einstein

Increments. The *increment* Δx of a variable x is the change in x as it increases or decreases from one value $x = x_0$ to another value $x = x_1$ in its domain. Here, $\Delta x = x_1 - x_0$ and we may write $x_1 = x_0 + \Delta x$. If the variable x is given an increment Δx from $x = x_0$ (i.e., if x changes from $x = x_0$ to $x_1 = x_0 + \Delta x$) and a function $y = f(x)$ is thereby given an increment $\Delta y = f(x_0 + \Delta x) - f(x_0)$ from $y = f(x_0)$, then the quotient

$$\frac{\Delta y}{\Delta x} = \frac{\text{change in } y}{\text{change in } x},$$

is called the *average rate of change* of the function on the interval between $x = x_0$ and $x_1 = x_0 + \Delta x$.

Let $f(x)$ be defined at any point x_0 in (a, b) . The derivative of $f(x)$ at $x = x_0$ is defined as

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

if this limit exists. A function is called **differentiable** at a point $x = x_0$, if it has a derivative at that point, i.e., if $f'(x_0)$ exists. If we write $x = x_0 + h$, then $h = x - x_0$ and h approaches 0 if and only if x approaches x_0 . Therefore, an equivalent way of stating the definition of the derivative, is

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

Example: If $f(x) = x^3 - x$, find a formula for $f'(x)$.

Solution:

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[(x+h)^3 - (x+h)] - [x^3 - x]}{h} \\&= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x - h - x^3 + x}{h} \\&= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 - h}{h} \\&= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2 - 1) = 3x^2 - 1.\end{aligned}$$

Example: If $f(x) = \sqrt{x}$, find the derivative of f .

Solution:

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\&= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \\&= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} \\&= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\&= \frac{1}{\sqrt{x} + \sqrt{x}} = \frac{1}{2\sqrt{x}}.\end{aligned}$$

The derivative at x may be denoted by $f'(x)$, y' , $\frac{dy}{dx}$, $\frac{d}{dx}(f(x))$. The symbol $\frac{d}{dx}$ is called **differentiation operator** because it indicates the operation of **differentiation**. The process of finding derivatives of functions is called **differentiation**.

A function f is differentiable at x_0 if $f'(x_0)$ exists. It is **differentiable on an open interval** (a, b) [or (a, ∞) or $(-\infty, a)$ or $(-\infty, \infty)$], if it is differentiable at every number in the interval.

Example: Where is the function $f(x) = |x|$ differentiable?

Solution: If $x > 0$, then $|x| = x$ and we can choose h small enough that $x+h > 0$ and hence $|x+h| = x+h$. Therefore, for $x > 0$,

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h} \\&= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1,\end{aligned}$$

and so f is differentiable for any $x > 0$.

Similarly, for $x < 0$, we have $|x| = -x$ and h can be chosen small enough that $x + h < 0$ and so $|x + h| = -(x + h)$. Therefore, for $x < 0$,

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{|x + h| - |x|}{h} \\ &= \lim_{h \rightarrow 0} \frac{-(x + h) - (-x)}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1, \end{aligned}$$

and so f is differentiable for any $x < 0$.

For $x = 0$ we have to investigate

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(0 + h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{|0 + h| - |0|}{h} \quad (\text{if it exists}). \end{aligned}$$

Let's compute the left and right limits separately;

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{|0 + h| - |0|}{h} &= \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} 1 = 1. \\ \lim_{h \rightarrow 0^-} \frac{|0 + h| - |0|}{h} &= \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} (-1) = -1. \end{aligned}$$

Since these limits are different $f'(0)$ does not exist. Thus, f is differentiable at all x except 0.

5.1 Differentiation Techniques (Finding Derivatives)

1.

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

if this limit exists.

2. The derivative of any constant function is zero, i.e., $c' = 0$.

3. For any real number n ,

$$(x^n)' = nx^{n-1}.$$

When differentiating, results can be expressed in a number of ways. For example, (i) if $y = 3x^2$ then $\frac{dy}{dx} = 6x$, (ii) if $f(x) = 3x^2$ then $f'(x) = 6x$, (iii) the differential coefficient of $3x^2$ is $6x$.

For example, if $f(x) = \sqrt{x} = x^{\frac{1}{2}}$, then $f'(x) = \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$.

You Try It: Using the general rule, differentiate the following with respect to x :

$$(a) f(x) = 5x^7 \quad (b) f(x) = \frac{4}{x^2}.$$

4. For any constant c , $(cf(x))' = cf'(x)$. For example, $(5x^3)' = 5(x^3)' = 5(3x^2) = 15x^2$.

5. The derivative of a sum (difference) is the sum (difference) of the derivatives,

$$(f(x) \pm g(x))' = f'(x) \pm g'(x).$$

For example,

$$(3x^5 - 2x^2 + 1)' = (3x^5)' - (2x^2)' + 1' = 3(5x^4)' - 2(2x)' + 0 = 3(5x^4) - 2(2x) = 15x^4 - 4x.$$

6. Product Rule

$$(f(x)g(x))' = f(x)g'(x) + g(x)f'(x).$$

7. Quotient Rule

$$\left(\frac{f(x)}{g(x)}\right)' = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}.$$

8. **Parametric Equations.** If the coordinates (x, y) of a point P on a curve are given as functions $x = f(u)$ and $y = g(u)$ of a third variable or *parameter* u , the equations $x = f(u)$ and $y = g(u)$ are called *parametric equations of the curve*. For example, $x = \frac{1}{2}t$, $y = 4 - t^2$ or $x = \cos \theta$, $y = 4 \sin^2 \theta$.

The First Derivative $\frac{dy}{dx}$ is given by

$$\frac{dy}{dx} = \frac{dy/du}{dx/du}.$$

The Second Derivative $\frac{d^2y}{dx^2}$ is given by

$$\frac{d^2y}{dx^2} = \frac{d}{du} \left(\frac{dy}{dx} \right) \frac{du}{dx}.$$

Example: Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ given $x = \theta - \sin \theta$ and $y = 1 - \cos \theta$.

Solution: Note that $\frac{dx}{d\theta} = 1 - \cos \theta$ and $\frac{dy}{d\theta} = \sin \theta$, so

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\sin \theta}{1 - \cos \theta}.$$

Also

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{d\theta} \left(\frac{\sin \theta}{1 - \cos \theta} \right) \frac{d\theta}{dx} \\ &= \frac{\cos \theta - 1}{(1 - \cos \theta)^2} \cdot \frac{1}{1 - \cos \theta} = -\frac{1}{(1 - \cos \theta)^2}. \end{aligned}$$

Example: Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ given $x = e^t \cos t$ and $y = e^t \sin t$.

Solution: Note that $\frac{dx}{dt} = e^t(\cos t - \sin t)$ and $\frac{dy}{dt} = e^t(\sin t + \cos t)$, so

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\sin t + \cos t}{\cos t - \sin t}.$$

Also

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dt} \left(\frac{\sin t + \cos t}{\cos t - \sin t} \right) \frac{dt}{dx} \\ &= \frac{2}{(\cos t - \sin t)^2} \cdot \frac{1}{e^t(\cos t - \sin t)} = \frac{2}{e^t(\cos t - \sin t)^3}.\end{aligned}$$

Theorem 5.1.1. *If $f(x) = c$ is a constant function, then $f'(x) = 0$ for all real numbers x .*

Proof. Observe that $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \frac{c - c}{h} = 0$. □

Theorem 5.1.2 (Product Rule). *If f and g are both differentiable at x , then the product function fg is also differentiable at x and $(fg)'(x) = f(x)g'(x) + g(x)f'(x)$.*

Proof.

$$(fg)'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}.$$

Trick of adding and subtracting $f(x+h)g(x)$ to the numerator,

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h} &= \\ \lim_{h \rightarrow 0} f(x+h) \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} + g(x) \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} &= \\ f(x)g'(x) + g(x)f'(x). &\end{aligned}$$

□

Theorem 5.1.3. *If g is differentiable at x and $g(x) \neq 0$, then*

$$\left(\frac{1}{g} \right)'(x) = -\frac{g'(x)}{(g(x))^2}.$$

Proof.

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{\frac{1}{g(x+h)} - \frac{1}{g(x)}}{h} &= \lim_{h \rightarrow 0} \frac{g(x) - g(x+h)}{h(g(x))(g(x+h))} \\ &= \lim_{h \rightarrow 0} \frac{-(g(x+h) - g(x))}{h} \frac{1}{g(x)g(x+h)} \\ &= \lim_{h \rightarrow 0} \frac{-(g(x+h) - g(x))}{h} \lim_{h \rightarrow 0} \frac{1}{g(x)g(x+h)} \\ &= -g'(x) \frac{1}{(g(x))^2}.\end{aligned}$$

□

Theorem 5.1.4. If f and g are differentiable at x and $g(x) \neq 0$, then $\frac{f}{g}$ is differentiable at x and

$$\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}.$$

Proof. Since $\frac{f}{g} = f \left(\frac{1}{g}\right)$, we have

$$\begin{aligned} \left(\frac{f}{g}\right)'(x) &= \left(f \cdot \frac{1}{g}\right)'(x) \\ &= f'(x)\frac{1}{g(x)} + f(x)\left(\frac{1}{g}\right)'(x) \\ &= \frac{f'(x)}{g(x)} + f(x)\left(-\frac{g'(x)}{(g(x))^2}\right) \\ &= \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}. \end{aligned}$$

□

Example: $f(x) = \frac{x^2 - 1}{x^2 + 1}$, then $f'(x) = \frac{(x^2 + 1)(2x) - (x^2 - 1)(2x)}{(x^2 + 1)^2} = \frac{4x}{(x^2 + 1)^2}$.

Example: If $f(x) = \frac{x}{x^2 + 1}$, then $f'(x) = \frac{(x^2 + 1) - x(2x)}{(x^2 + 1)^2} = \frac{1 - x^2}{(x^2 + 1)^2}$.

Example: If $f(x) = \frac{1}{x}$, then $f'(x) = -\frac{1}{x^2}$.

Example: If $x = \cos t$ and $y = t \sin t$, find $\frac{dy}{dx}$.

Solution:

$$\frac{dy}{dx} = \frac{\frac{d}{dt}(t \sin t)}{\frac{d}{dt}(\cos t)} = \frac{\sin t + t \cos t}{-\sin t}.$$

5.2 Derivatives of Trigonometric Functions

Recall :

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = 0.$$

$$\begin{aligned}
(\sin x)' &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\
&= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\
&= \lim_{h \rightarrow 0} \frac{\sin x(\cos h - 1) + \cos x \sin h}{h} \\
&= \lim_{h \rightarrow 0} \left(-\sin x \frac{(1 - \cos h)}{h} + \left(\cos x \frac{\sin h}{h} \right) \right) \\
&= -\sin x(0) + \cos x(1).
\end{aligned}$$

Hence $(\sin x)' = \cos x$.

Example: Find $\frac{dy}{dx}$ if $y = x^3 \sin x$.

Solution: $\frac{dy}{dx} = (x^3 \sin x)' = x^3(\sin x)' + \sin x(x^3)' = x^3 \cos x + 3x^2 \sin x$.

Example: Determine whether the function,

$$f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0, \end{cases}$$

is differentiable at $x = 0$.

Solution: Observe that

$$\lim_{h \rightarrow 0} \frac{h^3 \sin \frac{1}{h} - 0}{h} = \lim_{h \rightarrow 0} h^2 \sin \frac{1}{h} = 0.$$

Logarithmic Functions. Assume $a > 0$ and $a \neq 1$. If $a^y = x$, then define $y = \log_a x$. Let $\ln x \equiv \log_e x$ ($\ln x$ is called the *natural logarithm* of x).

Basic Properties of Logarithms

1. $\log_a 1 = 0$ (In particular, $\ln 1 = 0$).
2. $\log_a a = 1$ (In particular, $\ln e = 1$).
3. $\log_a uv = \log_a u + \log_a v$.
4. $\log_a \left(\frac{u}{v}\right) = \log_a u - \log_a v$.
5. $\log_a u^r = r \log_a u$.

Derivatives of $\ln x$ and e^x are $\frac{d}{dx} e^x = e^x$ and $\frac{d}{dx} \ln x = \frac{1}{x}$. Also $\frac{d}{dx}(a^x) = a^x \ln a$, $a > 0$.

Example: Calculate the derivative $\frac{d}{dx}[(\sin x + x) \cdot (x^3 - \ln x)]$.

Solution: We know that $\frac{d}{dx} \sin x = \cos x$, $\frac{d}{dx} x = 1$, $\frac{d}{dx} x^3 = 3x^2$ and $\frac{d}{dx} \ln x = \frac{1}{x}$. Therefore, by the addition rule,

$$\frac{d}{dx}(\sin x + x) = \frac{d}{dx} \sin x + \frac{d}{dx} x = \cos x + 1$$

and

$$\frac{d}{dx}(x^3 - \ln x) = \frac{d}{dx} x^3 - \frac{d}{dx} \ln x = 3x^2 - \frac{1}{x}.$$

Now we may conclude the calculation by applying the product rule;

$$\begin{aligned} \frac{d}{dx}[(\sin x + x) \cdot (x^3 - \ln x)] &= \frac{d}{dx}(\sin x + x) \cdot (x^3 - \ln x) + (\sin x + x) \cdot \frac{d}{dx}(x^3 - \ln x) \\ &= (\cos x + 1) \cdot (x^3 - \ln x) + (\sin x + x) \cdot \left(3x^2 - \frac{1}{x}\right) \\ &= 4x^3 - 1 + x^3 \cos x + 3x^2 \sin x - \frac{1}{x} \sin x - \ln x \cos x - \ln x. \end{aligned}$$

You Try It: Calculate the derivative

$$\frac{d}{dx} \left(\sin x \cdot \left(\cos x - \frac{x}{e^x + \ln x} \right) \right).$$

5.3 Derivative of a Composition [The Chain Rule]

We calculate the derivative of a composition by

$$[f \circ g(x)]' = f'(g(x)) \cdot g'(x).$$

If $y = f(u)$ where $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(u) \frac{du}{dx} = f'(g(x)) g'(x).$$

Similarly, if $y = f(u)$ where $u = g(v)$ and $v = h(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}.$$

Example: Calculate the derivative $\frac{d}{dx}(\sin(x^3 - x^2))$.

Solution: This is the composition of functions, so we must apply the Chain Rule. It is essential to recognize what function will play the role of f and what function will play the role of g . Notice

that, if x is the variable, then $x^3 - x^2$ is applied first and $\sin$ applied next. So it must be that $g(x) = x^3 - x^2$ and $f(s) = \sin s$. Notice that $\frac{d}{ds}f(s) = \cos s$ and $\frac{d}{dx}g(x) = 3x^2 - 2x$. Then

$$\sin(x^3 - x^2) = f \circ g(x)$$

and

$$\begin{aligned} \frac{d}{dx}(\sin(x^3 - x^2)) &= \frac{d}{dx}(f \circ g(x)) \\ &= \left[\frac{df}{ds}(g(x)) \right] \cdot \frac{d}{dx}g(x) \\ &= \cos(g(x)) \cdot (3x^2 - 2x) \\ &= [\cos(x^3 - x^2)] \cdot (3x^2 - 2x). \end{aligned}$$

Example: Calculate the derivative $\frac{d}{dx} \ln \left(\frac{x^2}{x-2} \right)$.

Solution: Let $h(x) = \ln \left(\frac{x^2}{x-2} \right)$. Then $h = f \circ g$, where $f(s) = \ln s$ and $g(x) = \frac{x^2}{x-2}$. So $\frac{d}{ds}f(s) = \frac{1}{s}$ and $\frac{d}{dx}g(x) = \frac{(x-2) \cdot 2x - x^2 \cdot 1}{(x-2)^2} = \frac{x^2 - 4x}{(x-2)^2}$. As a result,

$$\begin{aligned} \frac{d}{dx}h(x) &= \frac{d}{dx}(f \circ g) \\ &= \left[\frac{df}{ds}(g(x)) \right] \cdot \frac{d}{dx}g(x) \\ &= \frac{1}{g(x)} \cdot \frac{x^2 - 4x}{(x-2)^2} \\ &= \frac{1}{\frac{x^2}{x-2}} \cdot \frac{x^2 - 4x}{(x-2)^2} \\ &= \frac{x-2}{x^2} \cdot \frac{x^2 - 4x}{(x-2)^2} \\ &= \frac{x-4}{x(x-2)}. \end{aligned}$$

You Try It: Calculate the derivative of $\tan(e^x - x)$.

5.4 Continuity and Differentiation

What is the relationship between continuity and differentiation? It appears that functions that have derivatives must be continuous.

Theorem 5.4.1. *If a function f is differentiable at a point x , then it is continuous at x .*

Proof. We want to show that f is continuous at x , i.e., $\lim_{t \rightarrow x} f(t) = f(x)$ or $\lim_{h \rightarrow 0} f(x+h) = f(x)$, where $h = t - x$. It will be sufficient to show that $\lim_{h \rightarrow 0} [f(x+h) - f(x)] = 0$.

Now,

$$\begin{aligned} \lim_{h \rightarrow 0} [f(x+h) - f(x)] &= \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right] h \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \lim_{h \rightarrow 0} h \\ &= f'(x) \cdot 0 \\ &= 0, \end{aligned}$$

because $f'(x)$ is finite. Thus f is continuous at x . □

Converse is false: For example, the function $f(x) = |x|$ is continuous at $x = 0$, but it is not differentiable there.

5.5 Higher Order Derivatives

If $f(x)$ is differentiable in an interval, its derivative is given by $f'(x)$, y' or $\frac{dy}{dx}$ where $y = f(x)$.

If $f'(x)$ is also differentiable in the interval, its derivative is denoted by $f''(x)$, y'' or $\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2}$.

Similarly, the n th derivative of $f(x)$, if it exists, is denoted by $f^{(n)}$, $y^{(n)}$ or $\frac{d^ny}{dx^n}$ where n is called the order of the derivative.

Example: Let $y = f(x) = \frac{1}{2}x^4 - 3x^2 + 1$.

Solution: Derivative $y' = f'(x) = \frac{d}{dx}(\frac{1}{2}x^4 - 3x^2 + 1) = 2x^3 - 6x$.

Second derivative $y'' = f''(x) = \frac{d^2y}{dx^2} = \frac{d}{dx}(2x^3 - 6x) = 6x^2 - 6$.

Third derivative $y''' = f'''(x) = \frac{d^3y}{dx^3} = \frac{d}{dx}(6x^2 - 6) = 12x$.

Fourth derivative $y^{(4)} = f^{(4)}(x) = \frac{d^4y}{dx^4} = \frac{d}{dx}(12x) = 12$.

5.6 Implicit Differentiation

Compare

1. $x^2 - y^3 = 3 \iff y = \sqrt[3]{x^2 - 3}$.
2. $x^2 + y^2 = 1 \iff y = \pm\sqrt{1 - x^2}$.
3. $x^3 + y^2 = 3xy \iff$???????

Implicit Functions. A function in which the dependent variable is expressed solely in terms of the independent variable x , namely $y = f(x)$, is said to be an **explicit function**, for example, $y = \frac{1}{2}x^3 - 1$. An equation $f(x, y) = 0$, on perhaps certain restricted ranges of the variables, is said to define y *implicitly* as a function of x .

Example: (a) The equation $xy + x - 2y - 1 = 0$, with $x \neq 2$, defines the function $y = \frac{1-x}{x-2}$.
(b) The equation $4x^2 + 9y^2 - 36 = 0$ defines the function $y = \frac{2}{3}\sqrt{9-x^2}$ when $|x| \leq 3$ and $y \geq 0$ and the function $y = -\frac{2}{3}\sqrt{9-x^2}$ when $|x| \leq 3$ and $y \leq 0$.

The derivative y' may be obtained by one of the following procedures:

1. Solve, when possible, for y and differentiate with respect to x .
2. Thinking of y as a function of x , differentiate both sides of the given equation with respect to x and solve the resulting relation for y' . This differentiation process is known as *implicit differentiation*.

Example: Find $\frac{dy}{dx}$ if $x^2 + y^2 = 4$.

Solution: We differentiate both sides of the equation

$$\begin{aligned}\frac{d}{dx}x^2 + \frac{d}{dx}y^2 &= \frac{d}{dx}4 \\ 2x + 2y\frac{dy}{dx} &= 0 \quad .\end{aligned}$$

Solving the derivative yields

$$\frac{dy}{dx} = -\frac{x}{y}.$$

Example: Find $\frac{d^2y}{dx^2}$ if $x^2 + y^2 = 4$.

Solution: From the above example, we already know that the first derivative is

$$\frac{dy}{dx} = -\frac{x}{y}.$$

Hence by the Quotient Rule

$$\begin{aligned}\frac{d^2y}{dx^2} &= -\frac{d}{dx} \left(\frac{x}{y} \right) \\ &= -\frac{y \cdot 1 - x \cdot \frac{dy}{dx}}{y^2} \\ &= -\frac{y - x \left(-\frac{x}{y} \right)}{y^2} \quad \boxed{\text{Substituting for } \frac{dy}{dx}} \\ &= -\frac{y^2 + x^2}{y^3}.\end{aligned}$$

Noting that $x^2 + y^2 = 4$ permits us to write the second derivative as

$$\frac{d^2y}{dx^2} = -\frac{4}{y^3}.$$

Example: Find $\frac{dy}{dx}$ if $\sin y = y \cos 2x$.

Solution:

$$\begin{aligned}\frac{d}{dx} \sin y &= \frac{d}{dx} y \cos 2x \\ \cos y \frac{dy}{dx} &= y(-\sin 2x \cdot 2) + \cos 2x \frac{dy}{dx} \\ (\cos y - \cos 2x) \frac{dy}{dx} &= -2y \sin 2x \\ \frac{dy}{dx} &= -\frac{2y \sin 2x}{\cos y - \cos 2x}.\end{aligned}$$

Example: Find y' , given $xy + x - 2y - 1 = 0$.

Solution: We have

$$x \frac{d}{dx}(y) + y \frac{d}{dx}(x) + \frac{d}{dx}(x) - 2 \frac{d}{dx}(y) - \frac{d}{dx}(1) = \frac{d}{dx}(0)$$

or $xy' + y + 1 - 2y' = 0$, then $y' = \frac{1+y}{2-x}$.

Example: Find y' , given $x^2y - xy^2 + x^2 + y^2 = 0$.

Solution:

$$\begin{aligned}\frac{d}{dx}(x^2y) - \frac{d}{dx}(xy^2) + \frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) &= 0 \\ x^2 \frac{d}{dx}(y) + y \frac{d}{dx}(x^2) - x \frac{d}{dx}(y^2) - y^2 \frac{d}{dx}(x) + \frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) &= 0.\end{aligned}$$

Hence, $x^2y' + 2xy - 2xyy' - y^2 + 2x + 2yy' = 0$ and $y' = \frac{y^2 - 2x - 2xy}{x^2 + 2y - 2xy}$.

Example: Find y' and y'' , given $x^2 - xy + y^2 = 3$.

Solution:

$$\frac{d}{dx}(x^2) - \frac{d}{dx}(xy) = \frac{d}{dx}(y^2) = 2x - xy' - y + 2yy' = 0. \quad \text{So} \quad y' = \frac{2x - y}{x - 2y}.$$

Then

$$\begin{aligned}y'' &= \frac{(x - 2y) \frac{d}{dx}(2x - y) - (2x - y) \frac{d}{dx}(x - 2y)}{(x - 2y)^2} = \frac{(x - 2y)(2 - y') - (2x - y)(1 - 2y')}{(x - 2y)^2} \\ &= \frac{3xy' - 3y}{(x - 2y)^2} = \frac{3x \left(\frac{2x - y}{x - 2y} \right) - 3y}{(x - 2y)^2} = \frac{6(x^2 - xy + y^2)}{(x - 2y)^2} \\ &= \frac{18}{(x - 2y)^2}.\end{aligned}$$

You Try It: Find y'' , given $x^3 - 3xy + y^3 = 1$.

5.7 Logarithmic Differentiation

Take natural logarithm ($\ln$) both sides, differentiate implicitly and solve for y' .

Example: Compute y' if $y = \frac{x^2 \sqrt[3]{7x - 14}}{(1 + x^2)^4}$.

Solution: $y = \frac{x^2 \sqrt[3]{7x-14}}{(1+x^2)^4} \Rightarrow \ln y = \ln \left(\frac{x^2 \sqrt[3]{7x-14}}{(1+x^2)^4} \right).$

$$\begin{aligned} \ln y &= 2 \ln x + \frac{1}{3} \ln(7x-14) - 4 \ln(1+x^2) \\ \frac{1}{y} y' &= 2 \left(\frac{1}{x} \right) + \frac{1}{3} \left(\frac{(7x-14)'}{7x-14} \right) - 4 \left(\frac{(1+x^2)'}{1+x^2} \right) \\ &= \frac{2}{x} + \frac{7}{3(7x-14)} - \frac{8x}{1+x^2} \\ y' &= y \left(\frac{2}{x} + \frac{7}{3(7x-14)} - \frac{8x}{1+x^2} \right) \\ &= \frac{x^2 \sqrt[3]{7x-14}}{(1+x^2)^4} \left(\frac{2}{x} + \frac{7}{3(7x-14)} - \frac{8x}{1+x^2} \right). \end{aligned}$$

5.8 Derivatives of Inverse Trigonometric Functions

If $x = \sin y$, the inverse function is written $y = \sin^{-1} x$ or $y = \arcsin x$. The inverse trigonometric functions are *multivalued* functions.

Example: Find the derivative of $y = \sin^{-1} x$.

Solution: Differentiate implicitly with respect to x . Then $\sin y = x$. Hence,

$$\begin{aligned} (\sin y)' &= x' \\ \cos y y' &= 1 \\ y' &= \frac{1}{\cos y} \\ y' &= \frac{1}{\sqrt{1 - \sin^2 y}} \\ &= \frac{1}{\sqrt{1 - x^2}}. \end{aligned}$$

Some Derivatives. $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}, \quad \frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}.$

You Try It: Show that the derivative of $\tan^{-1} x = \frac{1}{1+x^2}$.

5.9 Tutorial Questions

1. Show directly from definition that the derivative of $f(x) = x^3$ is $3x^2$.

2. Show from definition that $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$.
3. Determine whether each of the following functions is differentiable at $x = 0$.
- (i) $f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ (ii) $f(x) = x|x|$ (iii) $f(x) = \begin{cases} \frac{x - |x|}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$.
4. Show that $f(x) = \cos x$ is differentiable at any $x \in \mathbb{R}$ and that $f'(x) = -\sin x$.
5. Find $\frac{dy}{dx}$ for each of the following functions
- (i) $y = \sin^2(3x + \frac{\pi}{6})$ (ii) $y = (ax + b)^m(cx + d)^n$ (iii) $y = \cos(x^2 - 3x + 1)$
- (iv) $y = \frac{x}{1 + x^2}$ (v) $y = \sqrt{\frac{ax + b}{cx + d}}$ (vi) $e^{xy} + y \ln x = \cos 2x$ (vii) $y = \sec^{-1} x$
- (viii) $x = t \cos t$ and $y = t \sin t$ (ix) $xy = \sin(x + y)$ (x) $\frac{x}{y^2} + \frac{y^2}{x} = 5$
- (xi) $\cos(x - y) = ye^{xy}$ (xii) $x = \sec t$ and $y = \tan t$ (xiii) $xy^3 - 3x^2 = xy + 5$.
6. Use logarithmic differentiation to find derivatives of each of the following functions
- (i) $y = x^x, x > 0$ (ii) $y = x^{\ln x}, x > 0$ (iii) $y = \sqrt[3]{\frac{(x+1)(x+2)}{(x^2+1)(x^2+2)}}$
- (iii) $y = \sqrt{x+1}\sqrt{x+2}\sqrt[4]{x+3}$ (iv) $y(x) = y_1(x)y_2(x) \cdots y_n(x)$ (v) $y = x^{\sqrt{x}}, x > 0$
- (vi) $y = \frac{\sqrt[3]{x^4 + 6x^2(8x+3)^5}}{(2x^2 + 7)^{\frac{2}{3}}}$ (vii) $y = (x^2 + 1)^{(x+1)^x}, x > -1$ (viii) $y = x^x 2^x$
- (ix) $y = x^{x^x}$ (x) $y = x^{\cos x}$ (xi) $y = x^{(x+1)^{\ln x}}$ (xii) $y = (\ln x)^x$ (xiii) $y = \frac{(x^2 + 1)^x}{x^2}$.
7. Use implicit differentiation to find $\frac{dy}{dx}$.
- (i) $(y - 1)^2 = 4(x + 2)$ (ii) $(x^2 + y^2)^6 = x^3 - y^3$ (iii) $y^4 - y^2 = 10x - 3$ (iv) $y = \sin xy$
- (v) $\frac{x + y}{x - y} = x$ (vi) $xy = \sin(x + y)$ (vii) $x + y = \cos xy$ (viii) $x \sin y - y \cos x = 1$.
8. Find $\frac{dy}{dx}$ if $xy^2 + 4y^3 + 3x = 0$ at $(1, -1)$.
9. Find $\frac{d^2y}{dx^2}$.
- (i) $4y^3 = 6x^2 + 1$ (ii) $xy^4 = 5$ (iii) $x + y = \sin y$ (iv) $y^2 - x^2 = \tan 2x$ (v) $x^3 + y^3 = 27$.
10. Use implicit differentiation to find $\frac{dy}{dx}$. Then, solve for y explicitly in terms of x and differentiate. Show that the two answers are equivalent.
- (i) $x^3y = x + 1$ (ii) $y \sin x = x - 2y$.
11. If $y = \tan x$, prove that $y''' = 2(1 + y^2)(1 + 3y^2)$.
12. Given $y^2 \sin x + y = \tan^{-1} x$. find y' .
13. Find the first derivatives of the following functions
- (i) $y = \sin^{-1}(2x - 3)$.

(ii) $y = \tan^{-1} 3x^2$.

(iii) $y = x\sqrt{a^2 - x^2} + a^2 \sin^{-1} \left(\frac{x}{a} \right)$.

(iv) $y = x \csc^{-1} \left(\frac{1}{x} \right) + \sqrt{1 - x^2}$.

(v) $y = \frac{1}{ab} \tan^{-1} \left(\frac{b}{a} \tan x \right)$.

14. Find the first derivatives of the following functions

(i) $y = \ln(x^3 + 2)(x^2 + 3)$.

(ii) $y = \ln \left(\frac{x^4}{(3x - 4)^2} \right)$.

(iii) $y = x^2 3^x$.

(iv) $y = \ln(x + \sqrt{1 + x^2})$.

(v) $y = \ln(\ln \tan x)$.

(vi) $y = \tan^2 e^{3x}$.

15. If $y = x^2 e^x$, show that $y''' = (x^2 + 6x + 6)e^x$.

16. If $y = e^{-2x}(\sin 2x + \cos 2x)$. Show that $y'' + 4y' + 8y = 0$.

17. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$

(i) $x = 2 + t, y = 1 + t^2$.

(ii) $x = 2 \sin t, y = \cos 2t$.

(iii) $x = \cos^3 \theta, y = \sin^3 \theta$.

(iv) $x = a(\cos \phi + \phi \sin \phi), y = a(\sin \phi - \phi \cos \phi)$.

(v) $x = t + \frac{1}{t}, y = t + 1$.